Mmesoma Nwachukwu

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Git hub profile, LinkedIn

EDUCATION

Coventry University *Expected Graduation May 2026*

BSc Computer Science – *Predicted First Class Honors*

Courses: Programming and Algorithms, Computer Systems, Mathematics for Computer Science, Artificial Intelligence, Theory of Computation, Advanced Algorithms, Data Science, Software Engineering

SCHOLARSHIPS AND AWARDS

- **Year in Industry Student of the Year –** National Highways *(1 of 10 selected from 400 participants)*
- **Santander Universities Scholarship –** Santander Bank *(1 of 100 nationwide)*
- **AWS Machine Learning Scholarship –** Top 2% of 50,000+ DeepRacer applicants
- **Best Canadian Foundation Student –** 16 Plus Schools
- **Intern of the Year –** Paradigm Finance

PROFESSIONAL EXPERIENCE

Software Engineer **Projekt Rising** **Dec 2025-Present**

- Building AI solutions for small and medium businesses by analysing their workflows, identifying automation gaps, and shaping model requirements that fit real operational needs.
- Integrating models into client systems by creating APIs, connecting data sources, and setting up monitoring routines that track accuracy, latency, and user outcomes.
- Delivering contract projects by running model experiments, improving performance, and preparing deployments that clients can use in day-to-day work.

Software Engineer Intern **SalesAPE.ai** **Dec 2025-Present**

- Designed and implemented MVP AI prototypes to validate product feasibility as a member of the "hacksquads" by testing ideas, validating technical feasibility, and providing working demos for leadership.
- Improved model reliability through experiment design and evaluation pipelines by documenting experiments, improving model reliability, and preparing prototypes that move closer to production use.

Digital Services Year in Industry **National Highways** **July 2024-July 2025**

- Developed a knowledge graph with **Retrieval-Augmented Generation (RAG)** to extract insights from a large corpus of interconnected documents.
- Built and validated a Retrieval Augmented Generation knowledge graph to extract insights from large engineering document sets supporting infrastructure asset management.
- Developed a geospatial segmentation model using satellite and asset data to identify roadside habitats, improving infrastructure planning and environmental asset management workflows.

AI Trainer **Outlier AI** **July 2024-May 2025**

- Trained generative AI models for code generation through problem-solving and output evaluation.
- Created functional test cases and provided technical feedback to improve AI-generated code
- Applied software engineering across Python, JavaScript, and C++ to develop next-gen coding tools.

Founder **FaceRegg** **May-August 2023**

- Founded and built an image verification SaaS using convolutional neural networks for facial and image matching, delivering automated identity validation for client security integrations.
- Designed model training and inference pipelines using **TensorFlow** and **OpenCV**, optimizing accuracy and response latency for real world image inputs.
- Developed deployment ready APIs and integrated the system into client workflows, enabling automated verification and reducing manual review overhead.

Research Intern **Imperial College London** **June-July 2023**

- Collaborated in a cross-functional team to design sustainable solutions in energy and infrastructure.
- Prototyped data-driven technologies addressing climate resilience and scalability.
- Worked with researchers and industry mentors to assess impact and feasibility of innovations.

Software Engineering Intern **Paradigm Finance** **Sept-December 2021**

- Implementing and adhering to security best practices to safeguard sensitive financial data.
- Assisted with credit risk analysis by building tools to assess client portfolios and generate reports.
- Developed and maintained internal web applications.

PROJECTS

ExcelPoint WebApp

- Designed and implemented a full stack RAG learning platform using **Django, React, PostgreSQL, and LangChain**, enabling semantic search and context aware chat over uploaded educational documents using vector embeddings and sentence transformers.
- Built scalable AI processing pipelines with **Celery** and **Redis** for asynchronous document analysis, quiz generation, and adaptive learning workflows, integrated with **AWS S3** storage and JWT authenticated APIs.

Payment Authorisation System

- Built a payment authorization simulator modeling card network flow, issuer behavior, retry logic, and fraud feedback using Python and PostgreSQL.
- Developed issuer aware decision logic that adapts retry and authentication strategies based on historical approval patterns to improve authorization outcomes.
- Created an evaluation framework comparing baseline rules against adaptive strategies, measuring financial impact and transaction performance.

LEADERSHIP EXPERIENCE

Head of Engineering **Responsible AI Governance Initiative** **Aug 2025-Present**

- Leading the engineering efforts for a nonprofit focused on responsible AI development and governance.
- Managing a cross-functional technology team to build projects aligned with ethical AI standards.
- Overseeing architecture and development of technical projects

SKILLS

- **Languages:** Python, C++, JavaScript, HTML/CSS, R, Typescript, SQL,
- **Tools:** TensorFlow, PyTorch, NumPy, Pandas, Databricks, PostgreSQL, MongoDB, Git, VS Code, Jupyter, AWS, Knowledge Graphs, React,

CERTIFICATIONS

- CISCO Networking Academy: Introduction to Networks
- Coursera – Convolutional Neuro Networks

VOLUNTEERING

- **Coventry University Sports Society** – Football captain
- **Big Bang Fair Volunteer 2025**